Channel representations of noisy coherent detection in continuous-variable quantum key distribution: Additive noise versus attenuator

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We study the security of continuous-variable quantum key distribution (CV-QKD) under asymmetric coherent detection in an untrusted-noise scenario. Building on our previous POVM-based description of homodyne and double-homodyne receivers, we replace the additive-noise channel with attenuator channel. For both homodyne and double-homodyne detection, we derive the corresponding Gaussian-channel parameters and compare the resulting Holevo information and asymptotic secret fraction with those obtained from the additive-noise model.

Our analysis shows that, while both channel models under asymmetry yield degradaded protocol performance, the attenuator model gives less pessimistic security bounds.

In the double-homodyne case, due to the dependence on the squeezing parameter that describes ideal measurements, channel parameters and thereby informational quantities defining secure key rate are dependent on this squeezing parameter. The dependence of Holevo information on squeezing parameter is drastically different for the channel models, being concave for additive noise channel and convex for attenuator channel. It follows that for this type of measurement, choice of both squeezing parameter value and channel representation affects the resulting secure key rate.

I. INTRODUCTION

Homodyne and double-homodyne (heterodyne) detection are standard coherent-detection schemes in continuous-variable quantum optics and quantum information [1–3]. Homodyne detection measures a single field quadrature, whereas double-homodyne detection implements a joint Gaussian measurement of two conjugate quadratures. These schemes constitute the measurement stage in a broad class of continuous-variable quantum-communication protocols, including protocols for continuous-variable quantum key distribution (CV-QKD) [4–6], quantum state tomography [7, 8] and quantum state preparation [9].

A practical coherent receiver deviates from its idealized description because of nonunit photodetector efficiencies, electronic noise, beam-splitter imbalance, and mismatched responses of the detection branches. Such imperfections modify the conditional probability distribution of Bob's outcome for a given incoming optical state and must therefore be included in the measurement model used in a security analysis [10–14]. Both ideal and imperfect coherent measurements are naturally described by positive operator-valued measures (POVMs). In the Gaussian regime, receiver imperfections give rise to noisy Gaussian POVMs.

For the class of noisy Gaussian measurements considered here, the measurement statistics can be represented either by applying a Gaussian channel to the incoming state before an ideal measurement or, equivalently, by applying the dual channel to the ideal POVM [15, 16]. This representation is particularly useful in CV-QKD because it incorporates receiver imperfections into the same Gaussian formalism used to model propagation and to evaluate the information available to an eavesdropper. Two natural Gaussian channels are the additive-noise channel and the attenuator channel. The former adds a zero-mean Gaussian random displacement whose quadratures are uncorrelated with the input quadratures, whereas the latter mixes the input mode with an environmental mode at a beam splitter. The attenuator reduces to a pure-loss channel when the environmental mode is in the vacuum state [3, 17].

Equivalence at the level of measurement statistics does not, however, make different channels interchangeable in a security analysis. With suitable parameter choices and, where necessary, a deterministic invertible rescaling of the outcomes, distinct Gaussian channels may reproduce the same calibrated receiver statistics and hence the same Alice-Bob mutual information. Nevertheless, their complementary channels need not be equivalent, so the environmental systems available to an eavesdropper may differ. This distinction is relevant to both untrusted- and trusted-receiver

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models. In an untrusted-receiver model, the environmental output of the detector-channel realization is conservatively included in Eve's system. In a trusted-receiver model, the corresponding environmental modes remain inside Bob's laboratory and are inaccessible to Eve. Thus, the trust assignment can change the Holevo information even when the observed receiver statistics remain unchanged [17–21]. As has been highlighted in a recent work [22], Gaussian channel representation choice, while describing the same physical reality, will significantly alter security estimates.

Previous studies have established that receiver imbalance and other nonidealities can bias channel-parameter estimation, reduce the secret-key rate, and create security gaps when omitted from the measurement model [10, 12–14]. These results motivate an investigation into how the security estimate changes when the same asymmetry-induced POVM is realized through different Gaussian channels. This approach separates the experimentally calibrated measurement model from the additional physical assumptions introduced by a particular channel realization.

In our previous work [23], we derived the Gaussian POVMs that account for beam-splitter imbalance and photodetector efficiency mismatch, collectively referred to as *asymmetry effects*, in homodyne and double-homodyne receivers. We represented the resulting measurement noise through an additive noise channel. For double-homodyne detection, the ideal Gaussian POVM was represented by a family of pure squeezed-state seeds, with the admissible interval of squeezing parameters fixed by the positivity of the excess-noise covariance matrix. Within the untrusted-noise model, this nonuniqueness made the Holevo information dependent on the squeezing parameter. Although mathematically direct, the additive-noise realization is not tied to a specific optical coupling mechanism and is not the only Gaussian realization compatible with the same noisy POVM.

In the present work, we construct and analyze an attenuator channel of the same asymmetric homodyne and double-homodyne POVMs and compare it with the additive noise channel. The channel parameters are obtained by requiring the dual channel to map the ideal POVM onto the target noisy POVM. For homodyne detection, the environmental mode is chosen to be in the vacuum state, so the attenuator takes the form of a pure-loss channel. For double-homodyne detection, the generally anisotropic measurement noise is reproduced using a pure Gaussian environmental state with unequal quadrature variances. The channels considered here therefore serve as detector-channel realizations of a fixed noisy measurement rather than as alternative models of the physical communication link.

We compare the two channels within the Gaussian-modulated coherent-state (GMCS) protocol [24] with reverse reconciliation in the asymptotic regime and evaluate the asymptotic secret fraction under collective Gaussian attacks [25, 26]. The GMCS protocol provides a natural setting for a direct covariance-matrix comparison of the two detector-channel realizations. In the security model adopted here, receiver asymmetry is treated as untrusted; accordingly, the environmental output of the selected realization is assigned to Eve. Finite-size effects, composable-security corrections, active detector attacks, and uncertainties in equipment calibration are outside the scope of the present work.

This paper is organized as follows. Section II, we briefly recall the formalism behind POVMs of asymmetrical homodyne and double homodyne measurement schemes established in Ref. [23]. In Sec. III, we establish equivalence between attenuator channel model primarily used in this work and additive noise channel model used in Ref. [23], and derive the coefficients of those Gaussian channels. In Sec. IV, we derive informational quantities defining asymptotic secret fraction in the GMCS protocol for both channel models and both coherent measurements schemes and compare the results. Finally, in Sec. V, we provide a discussion of the results as well as concluding remarks.

II. IMPERFECT GAUSSIAN MEASUREMENTS

We begin with a brief discussion of asymmetrical coherent measurements, schematically depicted in Fig. 1. Asymmetry effects are induced by imbalance (unequal transmission and reflection coefficients) of the beamsplitters and unequal quantum efficiency of the detectors. For detailed derivations of the expressions in this section see Ref. [23].

Q symbol of the POVM describing noisy homodyne measurements (see Fig. 1a) is given by

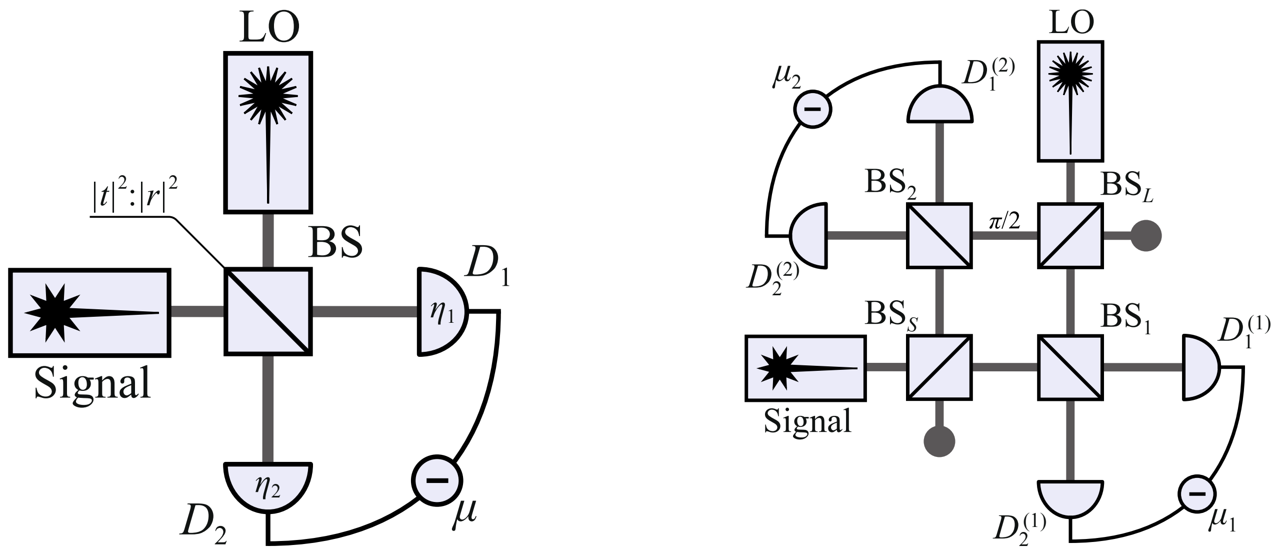
$$\mathsf{P}_G^{(\mathrm{H})}(x) = \frac{1}{\sqrt{2\pi\sigma_G}} \exp\left[-\frac{(x - \langle\hat{x}_\phi\rangle)^2}{2\sigma_x}\right], \tag{1}$$

$$\langle\hat{x}_\phi\rangle \equiv \langle\alpha|\hat{x}_\phi|\alpha\rangle = 2\,\mathrm{Re}\,\alpha e^{-i\phi}, \quad \phi = \arg\alpha_L, \tag{2}$$

$$\hat{x}_\phi = \hat{a}e^{-i\phi} + \hat{a}^\dagger e^{i\phi} \tag{3}$$

where α (α_L) is the complex amplitude of signal (LO) mode, $\hat{x}_\phi$ is the phase-rotated quadrature operator; x is the quadrature variable

$$x \equiv \frac{\mu}{(\eta_1+\eta_2)CS\,|\alpha_L|} - \frac{\eta_1 S^2 - \eta_2 C^2}{(\eta_1+\eta_2)\,CS}\,|\alpha_L| \tag{4}$$



(a) (b)

Figure 1: (a) Scheme of homodyne receiver. S (LO) is the source of signal (reference) mode, BS is the beamsplitter with transmission and reflection coefficients C and S, respectively; photodetectors D_1 and D_2 have quantum efficiencies η_1 and η_2, and $\mu = m_1 - m_2$ is the photon count difference. (b) Scheme of double homodyne receiver. S (LO) is the source of signal (reference) mode, BS$_S$ (BS$_L$) is the signal mode (local oscillator) beam splitter; $\frac{\pi}{2}$ is the quarter wave phase shifter; BS$_i$ is the beam splitter of ith homodyne; $D_{1,2}^{(i)}$ are the photodetectors of the ith homodyne; and $\mu_i = m_1^{(i)} - m_2^{(i)}$ is the photon count difference registered by the detectors of ith homodyne.

and σ_x is the quadrature variance

$$\sigma_x \equiv \frac{\eta_1 S^2 + \eta_2 C^2}{[(\eta_1+\eta_2)CS]^2}. \tag{5}$$

By computing inverse Forier transform of Q symbol (1), we obtain anti-normal ordered characteristic function [16]. From this characteristic function we can deduce the covariance matrix of the noisy homodyne measurements as follows

$$\Sigma_m^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}+\sigma_N, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{6}$$

$$\Sigma_{\mathrm{id}}^{(\mathrm{H})} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}, e^{2r})\,R^{\mathrm{T}}(\phi), \tag{7}$$

where $\Sigma_{\mathrm{id}}^{(\mathrm{H})}$ is the covariance matrix of ideal measurement, $R(\phi) = \begin{pmatrix} \cos\phi & -\sin\phi \\ \sin\phi & \cos\phi \end{pmatrix}$ is the rotation matrix; σ_N is the excess noise variance that accounts for asymmetry effects is given by

$$\sigma_N = \sigma_x - 1. \tag{8}$$

For double homodyne detection, Q symbol is given by

$$\mathsf{P}_G^{(\mathrm{DH})}(x_1,x_2) = \frac{1}{2\pi\sqrt{\sigma_G^{(1)}\sigma_G^{(2)}}}\exp\Big[-\frac{(x_1-\mathrm{Re}\,\alpha e^{-i\phi})^2}{\sigma_1} - \frac{(x_2-\mathrm{Im}\,\alpha e^{-i\phi})^2}{\sigma_2}\Big], \tag{9}$$

$$\sigma_G^{(i)} = (\eta_1^{(i)}S_i^2 + \eta_2^{(i)}C_i^2)|\alpha_L^{(i)}|^2, \quad \alpha_L^{(1)} = C_L\alpha_L, \quad \alpha_L^{(2)} = -iS_L\alpha_L, \tag{10}$$

where x_i are the quadrature variables given by

$$x_1 = \frac{1}{2(\eta_1^{(1)}+\eta_2^{(1)})C_1S_1C_S}\left\{\frac{\mu_1}{|\alpha_L^{(1)}|} - (\eta_1^{(1)}S_1^2 - \eta_2^{(1)}C_1^2)|\alpha_L^{(1)}|\right\}, \tag{11}$$

$$x_2 = \frac{1}{2(\eta_1^{(2)}+\eta_2^{(2)})C_2S_2S_S}\left\{\frac{\mu_2}{|\alpha_L^{(2)}|} - (\eta_1^{(2)}S_2^2 - \eta_2^{(2)}C_2^2)|\alpha_L^{(2)}|\right\}, \tag{12}$$

and relations

$$\sigma_1 = \frac{\sigma_x^{(1)}}{2C_S^2}, \quad \sigma_2 = \frac{\sigma_x^{(2)}}{2S_S^2}, \tag{13}$$

$$\sigma_x^{(i)} = \frac{\eta_1^{(i)} S_i^2 + \eta_2^{(i)} C_i^2}{\left[(\eta_1^{(i)} + \eta_2^{(i)}) C_i S_i\right]^2} \tag{14}$$

give the quadrature variances σ_1 and σ_2.

Following the same procedure used to derive Eq. (6), we obtain the covariance matrix of the double homodyne POVM,

$$\Sigma_m^{(\mathrm{DH})} = R(\phi)\,\mathrm{diag}(\delta_1, \delta_2) R^{\mathrm{T}}(\phi), \tag{15}$$

$$\delta_i = 2\sigma_i - 1. \tag{16}$$

The POVM that describes ideal double homodyne measurements uses the set of squeezed states,

$$|z, r, \phi\rangle = \hat{R}(\phi)\hat{D}(z)\hat{S}(r)|0\rangle = \hat{D}(ze^{i\phi})\hat{S}(re^{2i\phi})|0\rangle, \tag{17}$$

where z is the coherent state amplitude, r is real-valued squeezing parameter, and ϕ is squeezing phase, $\hat{D}(z) = e^{z\hat{a}^\dagger - z^*\hat{a}}$ is the displacement operator, and $\hat{S}(r) = e^{\frac{1}{2}\left(r\hat{a}^{\dagger 2} - r^*\hat{a}^2\right)}$ is the squeezing operator.

The covariance matrix of squeezed states reads

$$\Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} e^{2r} & 0 \\ 0 & e^{-2r} \end{pmatrix} R^{\mathrm{T}}(\phi). \tag{18}$$

An important point is that, for the difference between comvariance matrices of noisy and ideal double homodyne measurements, given by Eq. (15) and Eq. (18) respectively, is positive semi-definite, the squeezing parameter r must be ranged in the interval defined by parameters of the scheme (see Ref. [23])

$$r \in [r_2, r_1]\,, \; r_1 = \ln \delta_1^{1/2}, \; r_2 = \ln \delta_2^{-1/2}. \tag{19}$$

III. GAUSSIAN CHANNELS MODELING IMPERFECT MEASUREMENTS

For arbitrary state $\hat{\rho}$ and POVM $\hat{\Pi}^{(x)}$, the outcome probability is [27]

$$p(x|\hat{\rho}) = \mathrm{Tr}\left(\hat{\rho}\,\hat{\Pi}^{(x)}\right). \tag{20}$$

The measurement statistics of Gaussian states with noisy Gaussian measurements can be calculated by applying the noise map $\mathcal{C}$ to ρ_G, or the dual map $\mathcal{C}^*$ to the ideal POVM $\hat{\Pi}_{\mathrm{id}}$ [16], i.e.

$$p(x|\hat{\rho}_G) = \mathrm{Tr}\left\{\mathcal{C}(\hat{\rho}_G)\,\hat{\Pi}_{\mathrm{id}}^{(x)}\right\} = \mathrm{Tr}\left\{\hat{\rho}_G\,\mathcal{C}^*\left(\hat{\Pi}_{\mathrm{id}}^{(x)}\right)\right\}. \tag{21}$$

An important property is that the same measurement statistics can be described by different Gaussian channels $\mathcal{C}$, provided their dual maps $\mathcal{C}^*$ yield the same covariance matrix. While the first moments will differ, it is not as crucial due to being able to rescale them.

Note that, as measurements are specified to be Gaussian, the channels $\mathcal{C}$ and $\mathcal{C}^*$ will be Gaussian as well. It follows that the channel $\mathcal{C}$ is fully characterized by matrices $X_\mathcal{C}$ and $Y_\mathcal{C}$, transforming the first and second moments of Gaussian state ρ_G, $\boldsymbol{r}_G$ and Σ_G respectively, as follows [28]

$$\begin{aligned} &\boldsymbol{r}_G \mapsto X_\mathcal{C}\boldsymbol{r}_G, \\ &\Sigma_G \mapsto X_\mathcal{C}\Sigma_G X_\mathcal{C}^{\mathrm{T}} + Y_\mathcal{C}. \end{aligned} \tag{22}$$

The dual channel $\mathcal{C}^*$ transforming the first and second moments as follows

$$\begin{aligned} &\boldsymbol{r}_G \mapsto X_\mathcal{C}^{-1}\boldsymbol{r}_G, \\ &\Sigma_G \mapsto X_\mathcal{C}^{-1}\Sigma\left(X_\mathcal{C}^{-1}\right)^{\mathrm{T}} + X_\mathcal{C}^{-1}Y_\mathcal{C}\left(X_\mathcal{C}^{-1}\right)^{\mathrm{T}}. \end{aligned} \tag{23}$$

In what follows, we will consider two distinct quantum channels to describe the imperfect measurements, namely additive noise channel and attenuator channel. In accordance with Eq. (21), channel $\mathcal{C}$ will be defined in a way that its' dual channel $\mathcal{C}^*$ will transform the covariance matrix of ideal measurement Σ_{id} into the covariance matrix of noisy Gaussian measurement Σ_m,

$$\mathcal{C}^* : \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} \tag{24}$$

A. Additive noise channel

First, we consider additive noise (classical mixing) channel [16] Φ_γ, characterized by matrices

$$X_\Phi^{(\gamma)} = \mathbb{1}, \quad Y_\Phi^{(\gamma)} \geq 0, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{25}$$

This is a self-dual channel, $\Phi_\gamma^* = \Phi_\gamma$, that does not affect the first moment and changes the second moment as follows

$$\Phi_\gamma : \Sigma \mapsto \Sigma + Y_\Phi^{(\gamma)}. \tag{26}$$

To find matrices $Y_\Phi^{(\gamma)}$ so that (see Eq. (24))

$$\Phi_\gamma : \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \Sigma_{\mathrm{id}}^{(\gamma)} + Y_\Phi^{(\gamma)}, \tag{27}$$

we need to find the difference between covariance matrices of noisy and ideal measurements, i.e.

$$Y_\Phi^{(\gamma)} = \Sigma_m^{(\gamma)} - \Sigma_{\mathrm{id}}^{(\gamma)} \equiv \Sigma_N^{(\gamma)}, \tag{28}$$

where we defined the covariance matrix of excess noise $\Sigma_N^{(\gamma)}$.

For homodyne detection, from Eq. (7) and (6), we have

$$\Sigma_N^{(\mathrm{H})} = \Sigma_m^{(\mathrm{H})} - \Sigma_{\mathrm{id}}^{(\mathrm{H})} = R(\phi)\mathrm{diag}(\sigma_N, 0)R^{\mathrm{T}}(\phi) \geq 0, \tag{29}$$

where σ_N is given by Eq. (8).

For double homodyne detection, from Eq. (15) and (18), we have

$$\Sigma_N^{(\mathrm{DH})}(r) = \Sigma_m^{(\mathrm{DH})} - \Sigma_{\mathrm{id}}^{(\mathrm{DH})} = R(\phi) \begin{pmatrix} \sigma_N^{(1)}(r) & 0 \\ 0 & \sigma_N^{(2)}(r) \end{pmatrix} R^{\mathrm{T}}(\phi) \geq 0, \tag{30}$$

$$\sigma_N^{(1)}(r) = \delta_1 - e^{2r}, \quad \sigma_N^{(2)}(r) = \delta_2 - e^{-2r}. \tag{31}$$

Note that the elements of $\Sigma_N^{(\mathrm{DH})}$ are dependent on squeezing parameter that lies in the interval (19). When $r \in [r_2, r_1]$, it is certain that the covariance matrix of excess noise is positive semi-definite, $\Sigma_N^{(\mathrm{DH})}(r) \geq 0$.

B. Attenuator channel

We shall move on to the attenuator channel $\mathcal{E}_\gamma$, characterized by matrices [16]

$$X_{\mathcal{E}}^{(\gamma)} = \cos\theta_\gamma \mathbb{1}, \quad Y_{\mathcal{E}}^{(\gamma)} = \sin^2\theta_\gamma \Sigma_{\mathrm{env}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \tag{32}$$

where $\Sigma_{\mathrm{env}}^{(\gamma)}$ is the covariance matrix of the state decribing the enviroment.

Using Eq. (24) together with Eq. (22), we construct the dual channel $\mathcal{E}_N^{*(\gamma)}$ so that it transforms the POVM of the ideal measurement into the POVM of the noisy measurements by changing the covariance matrix $\Sigma_{\mathrm{id}}^{(\gamma)}$ into $\Sigma_m^{(\gamma)}$ as follows

$$\mathcal{E}_\gamma^* : \quad \Sigma_{\mathrm{id}}^{(\gamma)} \mapsto \Sigma_m^{(\gamma)} = \frac{1}{\cos^2\theta_\gamma} \Sigma_{\mathrm{id}}^{(\gamma)} + \tan^2\theta_\gamma \Sigma_{\mathrm{env}}^{(\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}. \tag{33}$$

First, we will consider homodyne detection. If the enviroment is decribed by vacuum, i.e.

$$\Sigma_{\rm env}^{(\rm H)} = \mathbb{1}, \tag{34}$$

together with from matrices of ideal (7) and noisy (6) measurements, Eq. (33) takes the form

$$\Sigma_m^{(\rm H)} = \lim_{r\to+\infty} R(\phi)\,\mathrm{diag}(e^{-2r}+\sigma_N, e^{2r})\,R^{\rm T}(\phi) = \lim_{r\to+\infty}\left\{\frac{1}{\cos^2\theta_{\rm H}} R(\phi)\,\mathrm{diag}(e^{-2r}, e^{2r})\,R^{\rm T}(\phi) + \tan^2\theta_{\rm H}\mathbb{1}\right\} \tag{35}$$

due to the limit $r\to\infty$, we may write

$$\frac{1}{\cos^2\theta}\Sigma_{\rm id}^{(\rm H)} \sim \Sigma_{\rm id}^{(\rm H)}, \quad e^{2r} + \mathrm{const} \sim e^{2r}, \tag{36}$$

it is sufficient to solve for $\tan^2\theta_{\rm H}$, i.e.

$$\tan^2\theta_{\rm H} = \sigma_N, \tag{37}$$

where σ_N is defined by Eq. (8). Then, $\cos^2\theta_{\rm H}$ and $\sin^2\theta_{\rm H}$ are given by

$$\cos^2\theta_{\rm H} = \frac{1}{\sigma_x}, \quad \sin^2\theta_{\rm H} = \frac{\sigma_N}{\sigma_x}. \tag{38}$$

For double homodyne detection, first, we note that, since $\Sigma_m^{(\rm DH)}$ is anisotropic, as generally quadrature variances $\delta_{1,2}$ are not equal, $\delta_1\neq\delta_2$, enviroment state must be anisotropic as well. Without loss of generality, we will consider the enviroment to be in pure state with $\det\Sigma_{\rm env}^{(\rm DH)} = 1$,

$$\Sigma_{\rm env}^{(\rm DH)} = R(\phi)\,\mathrm{diag}\left(a, \frac{1}{a}\right)\,R^{\rm T}(\phi), \quad a\geq 0. \tag{39}$$

Solving Eq. (33) for the case of double homodyne detection, (see Eq. (18), (15)),

$$\Sigma_m^{(\rm DH)} = R(\phi)\,\mathrm{diag}(\delta_1,\delta_2)R^{\rm T}(\phi) = R(\phi)\,\left\{\frac{1}{\cos^2\theta_{\rm DH}}\,\mathrm{diag}(e^{2r}, e^{-2r}) + \tan^2\theta_{\rm DH}\,\mathrm{diag}\left(a, \frac{1}{a}\right)\right\}\,R^{\rm T}(\phi), \tag{40}$$

$$\begin{cases}\delta_1 = \frac{e^{2r}}{\cos^2\theta_{\rm DH}} + \tan^2\theta_{\rm DH} a \\ \delta_2 = \frac{e^{-2r}}{\cos^2\theta_{\rm DH}} + \tan^2\theta_{\rm DH} a^{-1}\end{cases}. \tag{41}$$

Solving the system yields

$$\cos^2\theta_{\rm DH}(r) = \frac{e^{2r}+a(r)}{\delta_1+a(r)} = \frac{\delta_1 e^{-2r}+\delta_2 e^{2r}-2}{\delta_1\delta_2-1}, \tag{42}$$

$$a(r) = \frac{\delta_1 - e^{2r}}{\delta_2 e^{2r}-1}. \tag{43}$$

Note that at the ends of the r-interval (19), $r\to r_2$ or $r\to r_1$, either $a(r)$ or $1/a(r)$ tend to infinity, making the ancilla state (39) the eigenstate of the quadrature operator (3). By evaluating in this case the equations (41), i.e. resolving indeterminate form $\tan^2\theta_{\rm DH}a(r)$ for $r=r_2$ and $\tan^2\theta_{\rm DH}/a(r)$ for $r=r_1$, because one of noise variances (31) would be 0, it can be seen that the effect of this channel would be the same as the effect of the additive noise channel (27) modeling double homodyne detection with the same r. This fact will be used in numerical calculations (see next section and Fig. 3).

IV. GAUSSIAN MODULATED COHERENT STATES PROTOCOL

In the Gaussian modulated coherent states (GMCS) protocol, Alice sends coherent states $|\alpha = \alpha_1 + i\alpha_2\rangle$ drawn from the Gaussian prior

$$p_A(\alpha) = \frac{1}{\pi V_{\rm A}}\exp\left[-\frac{|\alpha|^2}{2V_{\rm A}}\right]. \tag{44}$$

The states pass the composite Gaussian channel $\mathcal{N}_{A\to B} = \mathcal{L}_N \circ \mathcal{L}_{T_{\rm ch}}$, where index indicates pure loss channel transmission coefficient, $T_{\rm ch}$ is the channel transmittance, $N = 1+\xi_{\rm ch}/(1-T_{\rm ch})$, and $\xi_{\rm ch}$ is the channel excess noise. Then, the receiver uses either homodyne or double homodyne detection to measure the states. Mutual information between legitimate parties in each case is then given by the following expressions (see Ref. [23] for derivation)

$$I_{\rm AB}^{\rm (H)} = \frac{1}{2}\log\left[1+\frac{4T_{\rm ch}V_{\rm A}}{\sigma_x+\xi_{\rm ch}}\right], \tag{45}$$

$$I_{\rm AB}^{\rm (DH)} = \frac{1}{2}\sum_i \log\left[1+\frac{4T_{\rm ch}V_{\rm A}}{2\sigma_i+\xi_{\rm ch}}\right], \tag{46}$$

where σ_x and σ_i given by Eqs. (5) and (14), respectively.

The covariance matrix of TMSV state with the Bob's mode transmitted through the noisy channel reads (see Ref. [29])

$$\mathcal{I}_A \otimes \mathcal{L}_N \circ \mathcal{L}_{T_{\rm ch}}: \quad \Sigma_{\rm TMSV} = \begin{pmatrix} V\mathbb{1} & \sqrt{V^2-1}\sigma_z \\ \sqrt{V^2-1}\sigma_z & V\mathbb{1}\end{pmatrix} \mapsto \Sigma_{\rm AB} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\rm B}\mathbb{1}\end{pmatrix}, \tag{47}$$

where $\mathcal{I}_A$ is the identity channel acting on the subsystem A and the parameters of the covariance matrix $\Sigma_{\rm AB}$ are given by

$$V = 1+4V_{\rm A}, \quad c = \sqrt{T_{\rm ch}(V^2-1)}, \tag{48}$$

$$V_{\rm B} = T_{\rm ch}(V-1)+1+\xi_{\rm ch}. \tag{49}$$

We consider local action of either attenuator channel $\mathcal{E}_\gamma$ (32) or additive noise channel Φ_γ (25), on second mode of TMSVS shared by Alice and Bob (47). In both cases, mutual information is given by $I_{\rm AB}^{(\gamma)}$ (45), (46), as it is classical.

It follows

$$\mathcal{I}_A \otimes \mathcal{E}_\gamma : \Sigma_{\rm AB} \mapsto \Sigma_{\rm AB}^{(\mathcal{E}_\gamma)} = \begin{pmatrix} V\mathbb{1} & c\cos\theta_\gamma\sigma_z \\ c\cos\theta_\gamma\sigma_z & \cos^2\theta_\gamma\left(V_{\rm B}+\tan^2\theta_\gamma\Sigma_{\rm env}^{(\gamma)}\right)\end{pmatrix}, \tag{50}$$

$$\mathcal{I}_A \otimes \Phi_N^{(\gamma)} : \Sigma_{\rm AB} \mapsto \Sigma_{\rm AB}^{(\Phi_\gamma)} = \begin{pmatrix} V\mathbb{1} & c\sigma_z \\ c\sigma_z & V_{\rm B}\mathbb{1}+\Sigma_N^{(\gamma)}\end{pmatrix}, \quad \gamma\in\{\rm H, DH\}. \tag{51}$$

When the channel $\mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}$ is under control of the eavesdropper, Eve holds a purification of the state $\hat{\rho}_{\rm AB}^{(\mathcal{C}_\gamma)} = \mathcal{I}_A \otimes \mathcal{C}_\gamma(\hat{\rho}_{\rm AB})$ and the measurements performed by the receiver are assumed to be noiseless. In this scenario, the difference between the joint and conditional entropies

$$\chi_{\rm EB}^{(\mathcal{C}_\gamma)} = S_{\rm E} - S_{\rm E|B} = S_{\rm AB}^{(\mathcal{C}_\gamma)} - S_{\rm A|B}^{(\mathcal{C}_\gamma)}, \quad \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}. \tag{52}$$

gives the Holevo information. The joint entropy

$$S_{\rm AB}^{(\mathcal{C}_\gamma)} = g(\nu_1^{(\mathcal{C}_\gamma)}) + g(\nu_2^{(\mathcal{C}_\gamma)}), \tag{53}$$

$$g(\nu) = \frac{\nu+1}{2}\log\frac{\nu+1}{2} - \frac{\nu-1}{2}\log\frac{\nu-1}{2}$$

then can be expressed in terms of the symplectic eigenvalues of the covariance matrix (51) given by

$$\nu_{1,2}^{(\mathcal{C}_\gamma)} = \sqrt{\Delta_{\mathcal{C}_\gamma}/2 \pm \sqrt{\Delta_{\mathcal{C}_\gamma}^2/4 - D_{\mathcal{C}_\gamma}}}, \quad D_{\mathcal{C}_\gamma} = \det\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}, \tag{54}$$

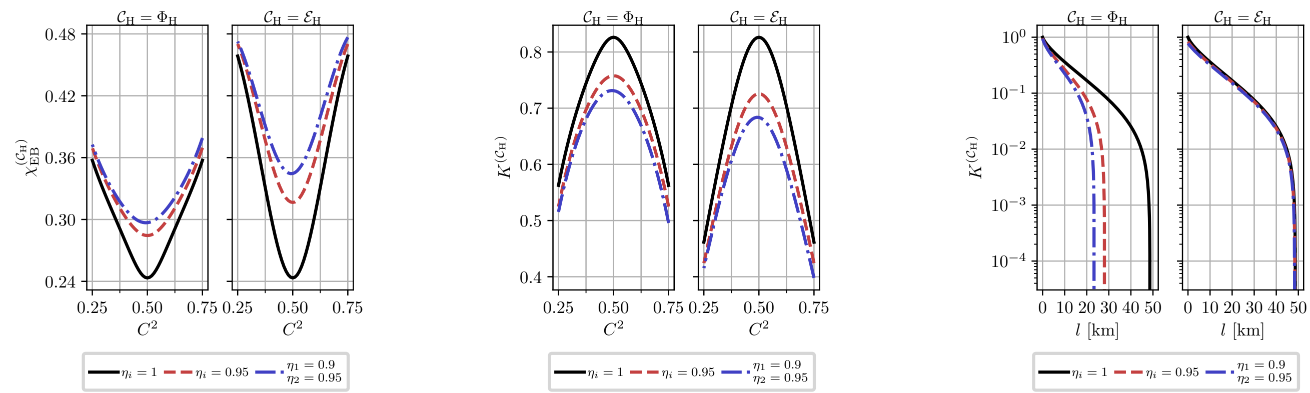
$$\Delta_{\mathcal{C}_\gamma} = \det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{11}) + \det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{22}) + 2\det([\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{12}), \tag{55}$$

where $[\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{ii}$ ($[\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}]_{ij}$) are diagonal (off-diagonal) blocks of $\Sigma_{\rm AB}^{(\mathcal{C}_\gamma)}$.

Conditional entropies $S_{\rm E|B} = S_{\rm A|B}^{(\gamma)}$ are calculated from respective conditional covariance matrices,

$$S_{\rm A|B}^{(\gamma)} = g(\nu_3^{(\gamma)}), \quad \nu_3^{(\gamma)} = \sqrt{\det\Sigma_{\rm A|B}^{(\gamma)}}, \tag{56}$$

$$\Sigma_{\rm A|B}^{(\gamma)} = V\mathbb{1} - c^2\sigma_z(V_{\rm B}\mathbb{1}+\Sigma_m^{(\gamma)})^{-1}\sigma_z. \tag{57}$$



(a) (b) (c)

Figure 2: (a) Holevo information, $\chi_{\mathrm{EB}}^{(C_{\mathrm{H}})}$ and (b) asymptotic secret fraction, $K^{(C_{\mathrm{H}})}$, as a function of the beam splitter transmission, C^2, of the homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5; T_{\mathrm{ch}} = 0.95; \xi_{\mathrm{ch}} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_{\mathrm{H}})}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

Direct computations show that these conditional matrices and therefore their symplectic eigenvalues are independent on channel model. Symplectic eigenvalues are given by

$$\nu_3^{(\mathrm{H})} = \sqrt{V\left(V - \frac{c^2}{V_{\mathrm{B}} + \sigma_N}\right)}, \tag{58}$$

$$\nu_3^{(\mathrm{DH})} = V\sqrt{\frac{\left(V_{\mathrm{B}} + \delta_1 - \frac{c^2}{V}\right)\left(V_{\mathrm{B}} + \delta_2 - \frac{c^2}{V}\right)}{\left(V_{\mathrm{B}} + \delta_1\right)\left(V_{\mathrm{B}} + \delta_2\right)}}, \tag{59}$$

From mutual information and Holevo information evaluation of the asymptotic secret fraction is done per Devetak-Winter formula [30]

$$K^{(\mathcal{C}_\gamma)} = \beta I_{AB}^{(\mathcal{C}_\gamma)} - \chi_{EB}^{(\mathcal{C}_\gamma)}, \quad \gamma \in \{\mathrm{H}, \mathrm{DH}\}, \mathcal{C}_\gamma \in \{\Phi_\gamma, \mathcal{E}_\gamma\}. \tag{60}$$

As symplectic eigenvalues for additive noise channel are calculated per formulas Eq. (54) and do not allow for simplification, in the following sections we focus only on analytic expressions for symplectic eigenvalues for attenuator channel model.

A. Homodyne detection

For attenuator channel, from Eq. (50) and (38) we have

$$\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})} = \begin{pmatrix} V\mathbb{1} & \frac{c}{\sqrt{\sigma_x}}\sigma_z \\ \frac{c}{\sqrt{\sigma_x}}\sigma_z & \frac{1}{\sigma_x}\left(V_{\mathrm{B}} + \sigma_N\right)\mathbb{1} \end{pmatrix} \equiv \begin{pmatrix} V\mathbb{1} & \tilde{c}\sigma_z \\ \tilde{c}\sigma_z & \tilde{V}_{\mathrm{B}}\mathbb{1} \end{pmatrix}, \tag{61}$$

$$\tilde{V}_{\mathrm{B}} \equiv T(V-1) + 1 + \xi, \tag{62}$$

$$\tilde{c} \equiv \sqrt{T(V^2 - 1)}, \tag{63}$$

where we defined scaled Bob's variance and scaled correlations between Alice and Bob $\tilde{V}_{\mathrm{B}}$ and $\tilde{c}$, using total channel transmission T that consists of channel transmission T_{ch} and detection losses $\frac{1}{\sigma_x}$, as well as scaled noise variance ξ,

$$T = T_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}, \tag{64a}$$

$$\xi \equiv \xi_{\mathrm{ch}} \cdot \frac{1}{\sigma_x}. \tag{64b}$$

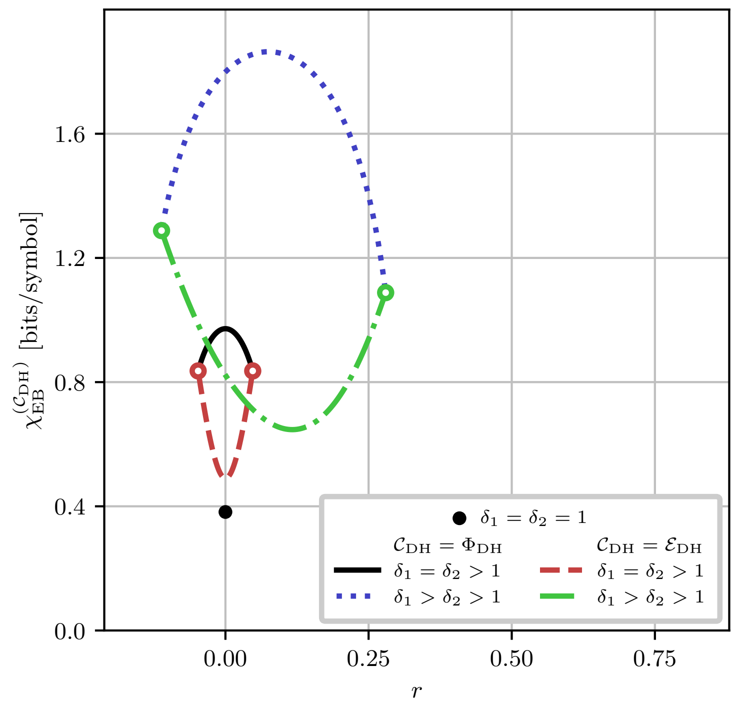


Figure 3: Holevo information, $\chi_{\mathrm{EB}}^{(\mathrm{DH};\mathcal{C}_{\mathrm{DH}})}$, as a function of the squeezing parameter r at different values of $\delta_{1,2}$. Black dot corresponds to the case of perfect measurements. Solid and dashdotted curves correspond to additive noise channel model, while dashed and dotted curves are for attenuator channel model. Color-matched white-filled dots mark points where Holevo information is undefined at the boundaries of the r interval (19) (see Eq. (43) and main text). Values of $\delta_{1,2}$ are: $\delta_{1,2} = 1.1$ for solid and dashed curves, $\delta_1 = 1.75, \delta_2 = 1.25$ for dashdotted and dotted curves. The channel parameters are: $V = 5$; $T_{\mathrm{ch}} = 0.95$; and $\xi_{\mathrm{ch}} = 10^{-2}$.

It follows that for homodyne detection the effect of measurement asymmetry is equivalent to scaling both channel transmission (64a) and channel noise (64b) by a factor of σ_x^{-1}.

Symplectic eigenvalues of $\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})}$, $\nu_{1,2}^{(\mathcal{E}_{\mathrm{H}})}$, can be computed using formulas [28]:

$$\nu_{1,2}^{(\mathcal{E}_{\mathrm{H}})} = \frac{z \pm [\tilde{V}_{\mathrm{B}} - V]}{2}, \quad z = \sqrt{\left(V + \tilde{V}_{\mathrm{B}}\right)^2 - 4\tilde{c}^2}, \tag{65}$$

and then substitued in Eq. (53) to calculate joint entropy $S_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{H}})}$.

Fig. 2 illustrates the effects of homodyne measurement asymmetry on Holevo information (52) and asymptotic secret fraction (60) versus beam splitter transmission C^2, as well as secret fraction versus transmission length, for both additive noise (51) and attenuator (50) channels. While channel noise is reduced in this setup (64b), so is channel transmission (64a), and, since transmission is scaled by modulation variance $V \gg 1$ in every expression, as well as reduced mutual information with greater asymmetry per Eq. (45), attenuator channel has negative effect. As shown in Fig. 2a and 2b, the attenuator channel results in higher Holevo information, and thus lower secret fraction, at high transmission for fixed asymmetry parameters. However, Fig. 2c reveals that the attenuator channel achieves significantly greater reach (maximum transmission length at which a positive key rate is obtained) than the additive noise channel.

B. Double homodyne

For attenuator channel, from Eq. (50) and (42) we have

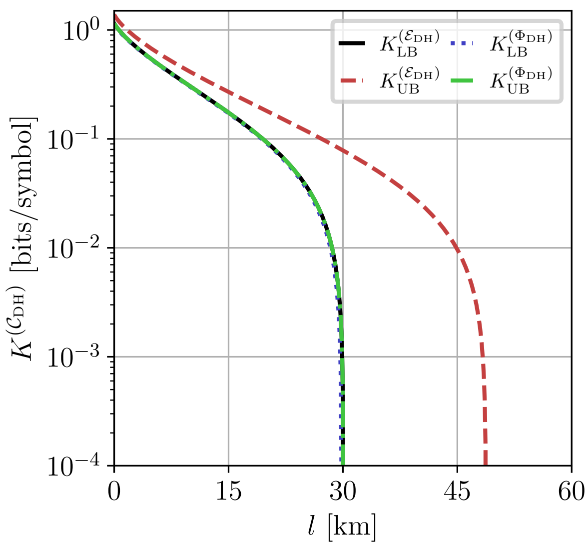
$$\Sigma_{\mathrm{AB}}^{(\mathcal{E}_{\mathrm{DH}})} = \begin{pmatrix} V\mathbb{1} & & c\cos\theta_{\mathrm{DH}}\sigma_z & \\ c\cos\theta_{\mathrm{DH}}\sigma_z & \cos^2\theta_{\mathrm{DH}}\left(V_{\mathrm{B}} + \delta_1 - \frac{e^{2r}}{\cos^2\theta_{\mathrm{DH}}}\right) & 0 \\ & 0 & \cos^2\theta_{\mathrm{DH}}\left(V_{\mathrm{B}} + \delta_2 - \frac{e^{-2r}}{\cos^2\theta_{\mathrm{DH}}}\right) \end{pmatrix} \tag{66}$$

Symplectic eigenvalues of this matrix can be calculated using invariants similarly to Eq. (54). It is obvious that these symplectic eigenvalues, and thereby joint entropy, is explicitly depended on squeezing parameter r (19), like in the case of additive noise channel (see Ref. [23]).

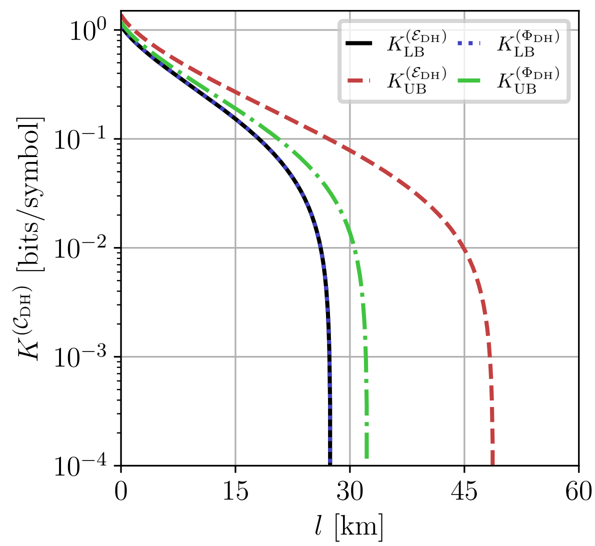
Fig. 3 illustrates dependence of Holevo information on the squeezing parameter, for both additive noise and attenuator channels. Referring to Fig. 3, in the limiting case of ideal double homodyne with $\delta_1 = \delta_2 = 1$, only one value of $r = 0$ is allowed (see Ref. [23]), with equal values of Holevo information for both channels (black dot on plot).

When $\delta_1 = \delta_2 > 1$, dependence of Holevo information on squeezing parameter is symmetrical around $r = 0$ for both channels. For additive noise channel, $r = 0$ is the point of maximal Holevo information, while for attenuator channel $r = 0$ corresponds to minimal Holevo information. Moreover, while Holevo information is unidefined at $r = r_{2,1}$ (see Eq. (42)), the values tend to the same values of joint entropy in the case of additive noise channel, as signified by color-matched white-filled dots in Fig. 3.

In asymmetrical case where $\delta_1 \neq \delta_2$, Holevo information for both channel models become asymmetrical relative to their extrema, resulting in unequal joint entropies at $r = r_{2,1}$. For additive noise channel, Holevo information is concave in r, while for attenuator channel it is convex in r.



(a) $q = 1$



(b) $q = 1.25$

Figure 4: Asymptotic secret fraction $K^{(C_{\mathrm{DH}})}$ versus channel length l, computed for fixed $\sigma_x^{(i)} = 1.05$ (see Eq.(14)) and various imbalance ratios $q \equiv \frac{S_S^2}{C_S^2}$ for additive noise and attenuator channel models (see Fig. 3). The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\mathrm{ch}} = 10^{-2}$; losses are 20 dB per 100 km.

Fig. 4 illustrates lower and upper bounds of the asymptotic secret fraction versus transmission length, for both additive noise (51) and attenuator (50) channels. Note that for chosen $\sigma_x^{(i)}$ lower bounds for both channels coincide, because for $\sigma_x^{(i)} \approx 1$ maximum of joint entropy for both models is either at $r = r_1$ or $r = r_2$. Moreover, the upper bound for additive noise channel in Fig. 4a is very close (0.2 km difference) to the lower bound.

As seen from Fig. 4, difference between lower and upper bound of the asymptotic secret fraction is greater in

Fig. 5 illustrates the dependence of Holevo information (52) and asymptotic secret fraction (60) on the signal beam splitter transmission C_S^2, as well as secret fraction versus transmission length if the receiver uses double homodyne detection, for both additive noise (51) and attenuator (50) channels. Unlike the homodyne detection case (see Fig. 2),

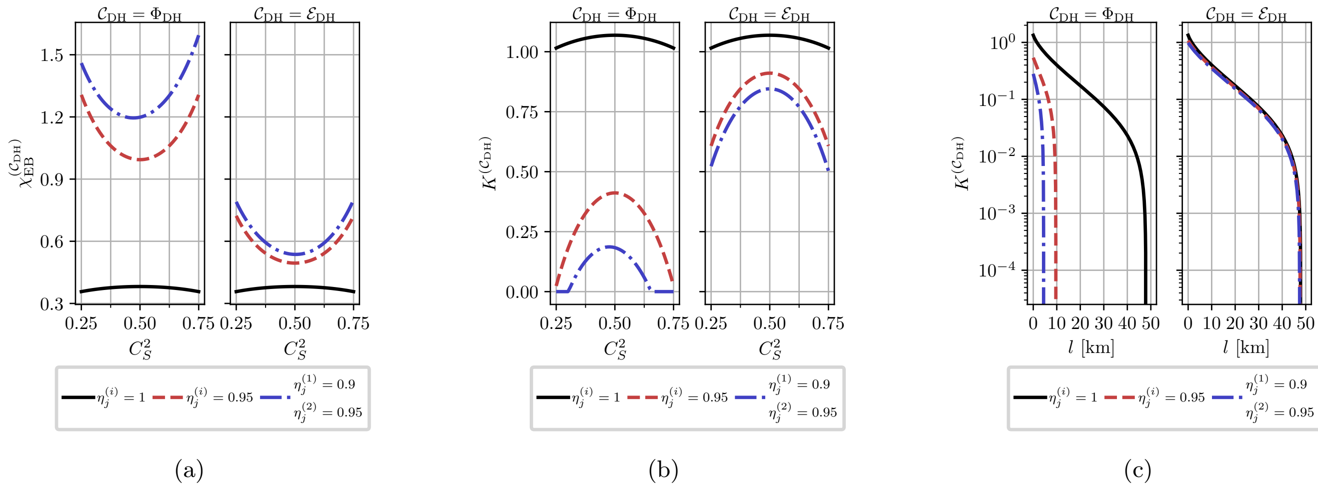


Figure 5: (a) Holevo information, $\chi_{\rm EB}^{(C_{\rm DH})}$ and (b) asymptotic secret fraction, $K^{(C_{\rm DH})}$, as a function of the signal beam splitter transmission, C_S^2, of the double homodyne receiver computed at different values of the detector efficiencies for additive noise and attenuator channel models, as indicated in panel title. The parameters are: $V = 5; T_{\rm ch} = 0.95; \xi_{\rm ch} = 10^{-2}$; and $\beta = 0.95$. (c) $K^{(C_{\rm DH})}$ versus transmission length l for additive noise and attenuator channel models. The parameters are: $V = 5$; $\beta = 0.95$; $\xi_{\rm ch} = 10^{-2}$; losses are 20 dB per 100 km.. For both channel models $r = \frac{r_2+r_1}{2}$ (see main text).

Holevo information is significantly lower for attenuator channel, except for the ideal measurements case ($\eta_{1,2}^{(1,2)} = 1$ and whole C_S^2 axis, unlike in homodyne case where ideal measurements are strictly $\eta_{1,2} = 1$ and $C^2 = 0.5$) as both channels have no effect and thereby have equal joint entropies (black curve). High difference in Holevo information is due to the dependence of joint entropy on r (see Fig. 3) and fixing $r = \frac{r_2+r_1}{2}$ for both curves, effectively minimizing Holevo information for attenuator channel and maximizing it for additive noise channel.

Analogously to homodyne detection case (see Fig. 2c), in double homodyne detection case the attenuator channel has greater reach than the additive noise channel, as illustated by Fig. 5c.

V. CONCLUSION

In this paper, we have studied two Gaussian channels that model measurement asymmetry in the untrusted-noise CV-QKD – the additive noise channel and the attenuator channel. Using the fact the action of the dual channel on the covariance matrix of the POVM of the ideal measurement must return the covariance matrix of the imperfect measurement (see Eq. (24)) to ensure that the measured statistics are equivalent (21), we have derived the coefficients that connect the models to the parameters of the scheme. This work directly addresses the methodological limitations of our previous study [23], where only the additive noise channel was considered.

Our comparative analysis reveals that the choice of the channel model significantly affects the evaluated bounds for the secure key rate of the GMCS protocol (see Fig. 2 and 5). While in both cases asymmetry is detrimental, the attenuator channel yields less pessimistic bounds and significantly greater reach (maximum transmission distance) that the additive noise channel. For homodyne detection, attenuator channel simply scales both the effective channel transmission and the excess noise by a factor of σ_x^{-1} (64).

For double-homodyne detection, an inherent ambiguity arises due to the dependence of the double homodyne POVM on the squeezing parameter, which we thoroughly discussed in Ref. [23]. Consequently, for both models, the channel coefficients depend on the squeezing parameter (19). Due to this dependency, the joint entropy, and therefore the Holevo information, depend on the squeezing parameter as well. Moreover, the dependence of the Holevo information on the squeezing parameter is drastically different for the two channel models (see Fig. 3). For additive channel, Holevo information is convex in r, whereas for attenuator channel it is concave in r.Due to this, the choices of both the channel model and the value of r yield significantly different results for the case of double-homodyne detection.

Our concluding remarks concern the generalization of our results. As shown in Appendix A

ACKNOWLEDGEMENTS

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Appendix A: Lossy beamsplitter

In this appendix, we show that applying independent pure loss channels at the output ports of a lossless beam splitter yields the exact same transformation as a single formally defined lossy beam splitter.

First, consider homodyne detection case depicted in Fig. ??.The lossless unbalanced beam splitter mixing the input signal mode $\hat{a}$ and the local oscillator mode $\hat{a}_L$. This transformation is described by the unitary rotation matrix $\mathbf{U}_{\mathrm{BS}}$,

$$\begin{pmatrix}\hat{a}_1\\ \hat{a}_2\end{pmatrix} = \mathbf{U}_{\mathrm{BS}}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} = \begin{pmatrix}C & -S\\ S & C\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} = \begin{pmatrix}C\hat{a}-S\hat{a}_L\\ S\hat{a}+C\hat{a}_L\end{pmatrix}, \tag{A1}$$

where C and S are the transmission and reflection coefficients of the beamsplitter BS, reslpectively.

We model the subsequent losses by passing each intermediate output mode $\hat{a}_{1,2}$ through a virtual beam splitter of transmittance $\tau_{1,2}$. This mixes each signal with an independent environmental vacuum mode $\hat{v}_{1,2}$, where $[\hat{v}_i, \hat{v}_j^\dagger] = \delta_{ij}$,

$$\begin{pmatrix}\hat{b}_1\\ \hat{c}_1\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_1} & -\sqrt{1-\tau_1}\\ \sqrt{1-\tau_1} & \sqrt{\tau_1}\end{pmatrix}\begin{pmatrix}\hat{a}_1\\ \hat{v}_1\end{pmatrix}, \tag{A2}$$

$$\begin{pmatrix}\hat{b}_2\\ \hat{c}_2\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_2} & -\sqrt{1-\tau_2}\\ \sqrt{1-\tau_2} & \sqrt{\tau_2}\end{pmatrix}\begin{pmatrix}\hat{a}_2\\ \hat{v}_2\end{pmatrix}. \tag{A3}$$

Tracing out the lost environmental modes $\hat{c}_{1,2}$, the remaining physical output modes $\hat{b}_{1,2}$ expand to

$$\begin{pmatrix}\hat{b}_1\\ \hat{b}_2\end{pmatrix} = \begin{pmatrix}\sqrt{\tau_1}C & -S\sqrt{\tau_1}\\ \sqrt{\tau_2}S & C\sqrt{\tau_2}\end{pmatrix}\begin{pmatrix}\hat{a}\\ \hat{a}_L\end{pmatrix} + \begin{pmatrix}-\sqrt{1-\tau_1}\hat{v}_1\\ \\ -\sqrt{1-\tau_2}\hat{v}_2\end{pmatrix}. \tag{A4}$$

From Eq. (A4), we extract the effective scattering matrix $\mathbf{T}_{\mathrm{BS}}$ and the fluctuation noise operators $\hat{F}_{1,2}$ representing the device excitations

$$\mathbf{T}_{\mathrm{BS}} = \begin{pmatrix}\sqrt{\tau_1}C & -S\sqrt{\tau_1}\\ \\ \sqrt{\tau_2}S & C\sqrt{\tau_2}\end{pmatrix}, \tag{A5}$$

$$\begin{pmatrix}\hat{F}_1\\ \\ \hat{F}_2\end{pmatrix} = \begin{pmatrix}-\sqrt{1-\tau_1}\hat{v}_1\\ \\ -\sqrt{1-\tau_2}\hat{v}_2\end{pmatrix}. \tag{A6}$$

It is not difficult to show that to preserve the Canonical Commutation Relations for the output fields, $[\hat{b}_i, \hat{b}_j^\dagger] = \delta_{ij}$, the noise operators $\hat{F}_{1,2}$ must satisfy

$$[\hat{F}_i, \hat{F}_j^\dagger] = \delta_{ij} - \sum_k T_{ik}T_{jk}^*, \tag{A7}$$

where T_{ik} are the elements of the scattering matrix $\mathbf{T}_{\mathrm{BS}}$ (A5). Evaluating this constraint explicitly using the elements of $\mathbf{T}_{\mathrm{BS}}$ yields:

$$[\hat{F}_1, \hat{F}_1^\dagger] = 1-\tau_1, \tag{A8}$$

$$[\hat{F}_2, \hat{F}_2^\dagger] = 1-\tau_2, \tag{A9}$$

$$[\hat{F}_1, \hat{F}_2^\dagger] = 0, \tag{A10}$$

which matches the commutators calculated from definitions (A6).

We now connect this setup to the formal, standard mathematical treatment of lossy beam splitters [31, 32]. This framework models loss via an interaction matrix $\mathbf{A}$ acting on ancillary modes, which must obey the positivity condition

$\mathbf{1}-\mathbf{A}\mathbf{A}^\dagger \geq 0$ to satisfy unitarity. Note that the extracted scattering matrix $\mathbf{T}_{\mathrm{BS}}$ (A5) can be factored into a product of a diagonal scaling matrix and the lossless unitary,

$$\mathbf{T}_{\mathrm{BS}} = \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \begin{pmatrix} C & -S \\ S & C \end{pmatrix} = \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix} \mathbf{U}_{\mathrm{BS}}. \tag{A11}$$

By defining the positive semi-definite matrix representing the loss properties as $\mathbf{A}\mathbf{A}^\dagger = \mathrm{diag}(1-\tau_1, 1-\tau_2)$, it follows that:

$$\sqrt{\mathbf{1}-\mathbf{A}\mathbf{A}^\dagger} = \begin{pmatrix} \sqrt{\tau_1} & 0 \\ 0 & \sqrt{\tau_2} \end{pmatrix}. \tag{A12}$$

Substituting this back into Eq. (A11) yields the polar decomposition of the scattering matrix of a lossy beam splitter in its generalized form [32]:

$$\mathbf{T}_{\mathrm{BS}} = \sqrt{\mathbf{1}-\mathbf{A}\mathbf{A}^\dagger}\, \mathbf{U}_{\mathrm{BS}}. \tag{A13}$$

Because the matrix $\mathbf{A}\mathbf{A}^\dagger$ representing the losses is diagonal, it is straightforward to see that to account for lossy beamsplitter in a homodyne setup it is sufficient to simply rescale the quantum efficiencies of detection: $\eta_{1,2} \to \sqrt{\tau_{1,2}}\eta_{1,2}$. The same argument would apply in the case of double homodyne detection.

We conclude this appendix by discussing the generalized (asymmetrical) beamsplitter model where input ports of the beamsplitter are characterized by different set of transmission and reflection coefficients. In this case, the non-unitary scattering matrix is written in the form [33]

$$\mathbf{S} = \begin{pmatrix} t & \rho \\ r & \tau \end{pmatrix}, \tag{A14}$$

where t, τ (r, ρ) are complex transmission (reflection) coefficients.While it can be factored into lossy and unitary parts similarly to (A13), this case cannot be covered by the formulas after some replacement, because it ruins quadrature representation:

$$|\alpha, \alpha_L\rangle \mapsto |\alpha_1, \alpha_2\rangle, \tag{A15}$$
$$\alpha_1 = t\alpha + \rho\alpha_L, \quad \alpha_2 = r\alpha + \tau\alpha_L, \tag{A16}$$

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{A17}$$
$$\sigma_G = \eta_1|\alpha_1|^2 + \eta_2|\alpha_2|^2 \approx (\eta_1\rho^2 + \eta_2\tau^2)\,|\alpha_L|^2 + \frac{\eta_1 t\rho + \eta_2\tau r}{2}\,|\alpha_L|\,\langle\hat{x}_\phi\rangle, \tag{A18}$$
$$\mu_G = \eta_1|\alpha_1|^2 - \eta_2|\alpha_2|^2 \approx (\eta_1\rho^2 + \eta_2\tau^2)\,|\alpha_L|^2 + \frac{\eta_1 t\rho - \eta_2\tau r}{2}\,|\alpha_L|\,\langle\hat{x}_\phi\rangle \tag{A19}$$

Appendix B: Fluctuating Local Oscillator

In this Appendix, we extend the theoretical treatment of homodyne measurements to account for classical intensity fluctuations of the local oscillator (LO).

We start from the Gaussian approximation of the photon count difference probability distribution in homodyne scheme [23]

$$\mathsf{P}_G(\mu) = G(\mu - \mu_G; \sigma_G), \tag{B1}$$
$$\sigma_G = \eta_1|\alpha_1|^2 + \eta_2|\alpha_2|^2 \approx (\eta_1 S^2 + \eta_2 C^2)\,|\alpha_L|^2, \tag{B2}$$
$$\mu_G = \eta_1|\alpha_1|^2 - \eta_2|\alpha_2|^2 \approx (\eta_1 S^2 - \eta_2 C^2)\,|\alpha_L|^2 + CS(\eta_1 + \eta_2)\,|\alpha_L|\,\langle\hat{x}_\phi\rangle, \tag{B3}$$

where

$$G(x;\sigma) \equiv \frac{1}{\sqrt{2\pi\sigma}} \exp\Bigl(-\frac{x^2}{2\sigma}\Bigr). \tag{B4}$$

We assume that the LO amplitude fluctuates around its mean value α_L such that

$$\alpha_L = \alpha_L + \delta\alpha_L, \quad \delta\alpha_L \sim G(\delta\alpha_L, \sigma_L) \tag{B5}$$

where σ_{LO} represents the variance of the LO amplitude fluctuations.

To find the total probability distribution $\mathsf{P}_{\mathrm{tot}}(\mu)$ that accounts for the local oscillator fluctuations, we perform an ensemble average by integrating the conditional distribution $\mathsf{P}_G(\mu)$ over the Gaussian prior of the LO amplitude.

$$\mathsf{P}_{\mathrm{total}}(\mu) = \int \mathrm{d}\delta\alpha_L\, \mathsf{P}_G(\mu|\delta\alpha_L)\mathsf{P}_G(\delta\alpha_L) = \tag{B6}$$

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